FALCULTY OF COMPUTING

SESSION: II 2025/2026

PROGRAM/CLASS:

SECJ3623

COURSE:

MOBILE APPLICATION PROGRAMMING

LECTURER:

AP. PROF.DR. MOHD SHAHIZAN OTHMAN

ASSESSMENT:

FINAL PROJECT REPORT

NAME DETAILS:

NAME	REG. NO
THEVENDRAN A/L PAREMESWARAN	SX220343ECJHS04
MUHAMMAD FAIZAL BIN ASARAB ALI	SX210500ECJHS04
KIGENDRAN SIVA KUMAR	SX171065CSJS04

DATE:

05 OCT 2025

1.0 Stakeholder and Users Identification

1.0 Stakeholder Overview

The Pizza Ordering App is designed to serve a variety of stakeholders with distinct roles and requirements. **Customers** are the primary end-users who browse the pizza menu, place orders, make payments, and track deliveries in real time. They expect a smooth and intuitive interface, accurate order details, timely notifications, and secure payment options to ensure a convenient and satisfying experience. Team member **Kigen** focused on creating an engaging and user-friendly interface for customers, ensuring that the app is easy to navigate and visually appealing.

Restaurant staff and admins manage the menu, update prices, handle promotions, and monitor incoming orders. They need an admin panel to efficiently add or remove menu items, manage stock availability, and oversee the order fulfillment process. Team member **Tvan** worked on building this admin panel, implementing features that allow staff to manage the menu dynamically and monitor all active orders in real time.

Delivery staff are responsible for picking up and delivering orders promptly. They require access to order details, delivery addresses, and notifications for new assignments. The app provides real-time tracking to ensure timely deliveries and efficient communication with the restaurant and customers. Tvan also contributed to the delivery module, making sure delivery personnel can update order statuses and track routes effectively.

The **app developers and technical team** ensure the app is robust, secure, and scalable. They are responsible for backend integration, state management, performance optimization, and maintaining data security. Team member **Fizal** handled backend development, integrating Firebase Authentication, Firestore, Storage, and implementing BLoC state management for predictable, real-time updates. This ensures smooth functionality and reliable communication between all stakeholders.

Finally, **business owners and support staff** oversee the operational and strategic aspects of the app. Business owners monitor analytics, order trends, and customer feedback, while support staff handle inquiries and complaints. Together, Kigen, Tvan, and Fizal collaborated to deliver a comprehensive platform that addresses the needs of customers, staff, delivery personnel, and management, ensuring a seamless and secure pizza ordering experience for all stakeholders

1.2 User Roles & Responsibilities

The Pizza Ordering App has multiple user roles, each with specific responsibilities to ensure smooth operation and a tailored experience.

1. Customer
    - Browse the pizza menu, select items, and customize orders.
    - Place orders, make secure payments, and track order status in real time.
    - Provide feedback or ratings for pizzas and delivery services.
    - Manage their profile, including contact information and delivery addresses.
    -
2. Restaurant Admin / Staff
    - Manage the pizza menu, including adding, updating, or removing menu items.
    - Set prices, promotions, and special deals for customers.
    - Monitor incoming orders, update order statuses, and ensure timely fulfillment.
    - Access reports and analytics on order trends, popular items, and inventory levels.
    -
3. Delivery Staff
    - Receive and accept delivery assignments from the app.
    - Access customer addresses, order details, and delivery instructions.
    - Update delivery status in real time (e.g., picked up, on the way, delivered).
    - Communicate with customers if needed to ensure timely delivery.
    -
4. Business Owner
    - Monitor overall app performance, including sales, revenue, and customer engagement.
    - Analyze order trends and customer feedback to make strategic business decisions.
    - Manage administrative privileges, assign roles, and oversee staff performance.
    -
5. Support Staff
    - Handle customer inquiries, complaints, and technical issues.
    - Ensure timely response to feedback and resolve problems efficiently.
    - Coordinate with admins and delivery staff to address operational issues

2.0 Use Case Analysis

2.1 Actor Identification

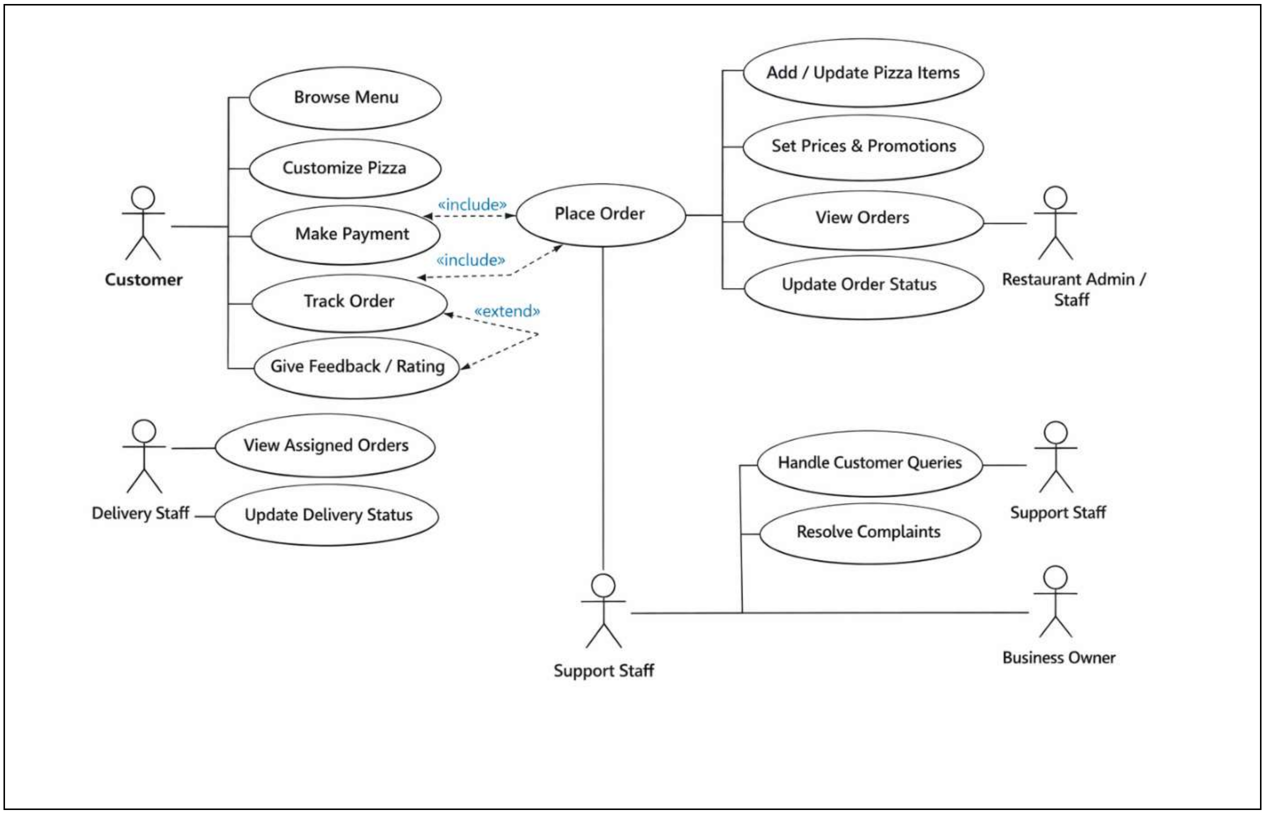
Actor identification is an important step in developing the Pizza Ordering App as it helps determine who interacts with the system and what functions they perform. Actors represent external users or entities that communicate with the application to achieve specific goals.

The primary actor is the **Customer**, who uses the app to browse the pizza menu, customize orders, make payments, track deliveries, and provide feedback. Customers interact directly with most front-end features, making them the central user of the system. The **Restaurant Admin/Staff** act as system managers responsible for maintaining menu items, updating prices, managing promotions, and processing incoming orders. They ensure that customer orders are prepared and updated in the system in a timely manner.

Another key actor is the **Delivery Staff**, who handle the logistics of delivering orders. They receive assigned deliveries, access customer addresses, and update delivery statuses such as “Picked Up,” “On the Way,” and “Delivered.” The **Support Staff** are responsible for handling customer inquiries, complaints, and service issues. They interact with the system to view orders and assist customers when problems arise.

Finally, the **Business Owner** is an external stakeholder who monitors business performance through reports and analytics. They review revenue, order trends, and customer feedback to make strategic and operational decisions.

2.2 UML Use Case Diagram



3.0 System Flow and Navigation Structure

3.1 Overall System Flow

The overall system flow of the Pizza Ordering App describes how users interact with the system from login to order completion. The process begins when a customer launches the application and logs in or registers using Firebase Authentication. Once authenticated, the customer is directed to the home screen, where they can browse the pizza menu, view promotions, and select items.

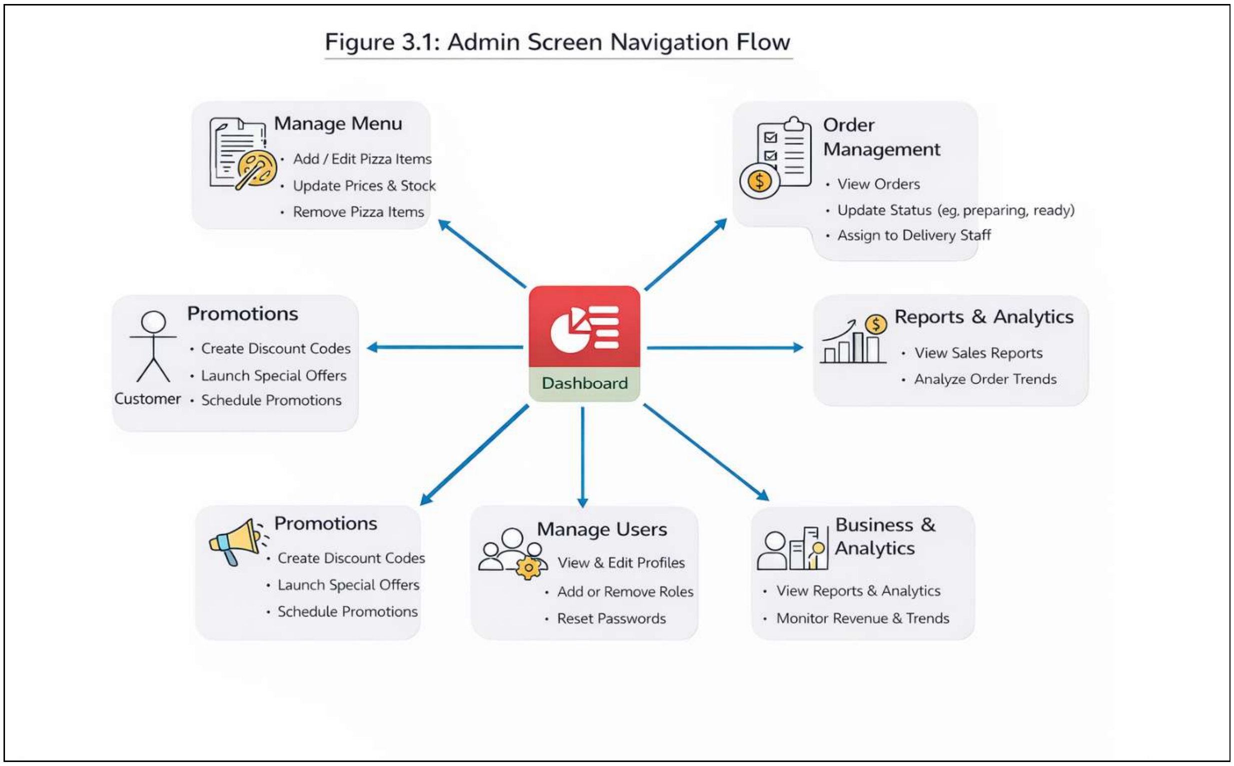
After selecting pizzas, the customer can customize the order (size, toppings, quantity) and add items to the cart. The system calculates the total price automatically. The customer then proceeds to checkout, confirms the delivery address, and completes payment through the integrated payment gateway. Once payment is successful, the system creates an order record in Firestore and sends a notification to the restaurant admin.

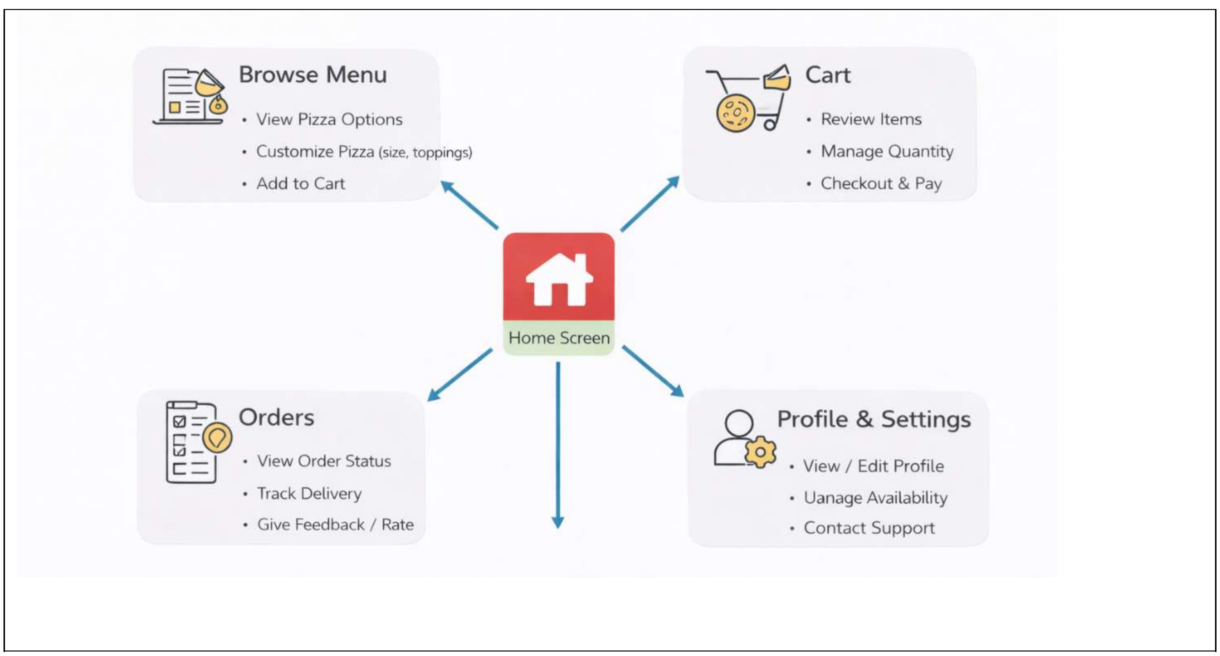
Next, the restaurant admin/staff receive the order details in real time. They review the order, prepare the pizzas, and update the order status (e.g., “Preparing,” “Ready for Pickup”). When the order is ready, it is assigned to available delivery staff through the system. The delivery staff receive a notification with delivery details, including customer address and contact information. They pick up the order, update the delivery status (e.g., “On the Way”), and deliver the pizza to the customer. Real-time tracking allows customers to monitor delivery progress within the app.

Finally, once the order is delivered, the system updates the status to “Completed.” Customers can provide ratings and feedback, while business owners can analyze order data and performance metrics through the admin dashboard. This end-to-end flow ensures efficient coordination between customers, restaurant staff, and delivery personnel.

3.2 User Navigation Structure

The following is a Navigation diagram for Pizza Ordering App





4.0 User Interface Design

The User Interface (UI) design of the Pizza Ordering App focuses on providing a simple, attractive, and user-friendly experience for all users, especially customers. The design follows modern mobile application standards with intuitive navigation, clear layouts, and visually appealing components. The main objective is to ensure users can browse menus, place orders, and track deliveries with minimal effort.

The home screen acts as the central navigation hub, displaying featured pizzas, promotions, categories, and quick access icons to Menu, Cart, Orders, and Profile. A clean card-based layout is used to showcase pizza items with images, prices, and ratings, making selection easier and more engaging.The menu screen allows customers to browse available pizzas and customize them based on size, crust type, toppings, and quantity. Filters and search functions help users quickly find their preferred items. Each pizza detail page includes images, descriptions, ingredients, and pricing information.

The cart and checkout interface is designed for clarity and efficiency. Users can review selected items, update quantities, remove pizzas, and view the total price calculation. The checkout screen collects delivery address details and integrates secure payment options before confirming the order. For order management, the orders screen displays real-time order status such as Preparing, On the Way, and Delivered. Customers can track delivery progress and provide ratings or feedback after completion. Meanwhile, profile and settings screens allow users to manage personal information, addresses, and app preferences.

Overall, the UI design emphasizes responsiveness, accessibility, and consistency across all screens, ensuring a smooth and enjoyable pizza ordering experience for customers, staff, and administrators

4.1 Low Fidelity UI Design

Low-fidelity UI design represents the early visual structure of the Pizza Ordering App. It focuses on layout, navigation flow, and feature placement rather than colors, typography, or detailed graphics. These wireframes help the development team and stakeholders quickly understand how screens are organized and how users interact with the system before moving to high-fidelity designs.

The **Home Screen wireframe** typically contains a top navigation bar (logo, search, profile icon), promotional banner area, pizza categories, and a scrollable list of pizza cards. Each card shows a placeholder image box, pizza name, price, and an “Add to Cart” button. Bottom navigation includes Menu, Cart, Orders, and Profile for quick access.

The **Menu & Pizza Details wireframes** focus on browsing and customization. The menu screen includes category filters, search bar, and a vertical list/grid of pizzas.

Selecting an item opens the pizza details screen, showing image placeholders, size options, toppings checkboxes, quantity selector, and “Add to Cart” button.

The **Cart & Checkout wireframes** illustrate the order review and payment flow. The cart screen lists selected pizzas with quantity controls, price totals, and a checkout button. The checkout screen contains delivery address fields, payment method selection, and order confirmation.

The **Order Tracking & Profile wireframes** show post-purchase interactions. The order tracking screen displays order status steps (Preparing → Out for Delivery → Delivered) and delivery details. The profile screen includes user information, saved addresses, order history, and logout options.

4.2 High Fidelity UI Design

High-fidelity UI design represents the final visual appearance of the Pizza Ordering App. Unlike low-fidelity wireframes, high-fidelity screens include real colors, typography, images, icons, branding elements, and interactive components. These designs closely resemble the actual mobile application and are used for usability testing, stakeholder approval, and developer handoff.

The overall design follows a **modern food delivery theme** using appetizing pizza imagery, warm color palettes (reds, oranges, yellows), clean typography, and card-based layouts. Consistent navigation patterns such as bottom navigation bars and floating action buttons ensure users can easily move between Menu, Cart, Orders, and Profile screens.

1. Home Screen (Dashboard UI)

The home screen highlights featured pizzas, promotions, and quick categories. High-resolution pizza images, discount banners, ratings, and pricing are displayed in visually engaging cards to encourage ordering.

2. Menu & Pizza Details Screen

The menu screen displays a categorized pizza list with filters, search bar, and product cards. The pizza details screen includes large images, ingredient lists, size selection, toppings customization, quantity selector, and add-to-cart button.

3. Cart & Checkout Screen

This interface allows users to review selected items, adjust quantities, view price breakdowns, and proceed to secure payment. Delivery address input and payment options (card, e-wallet) are included.

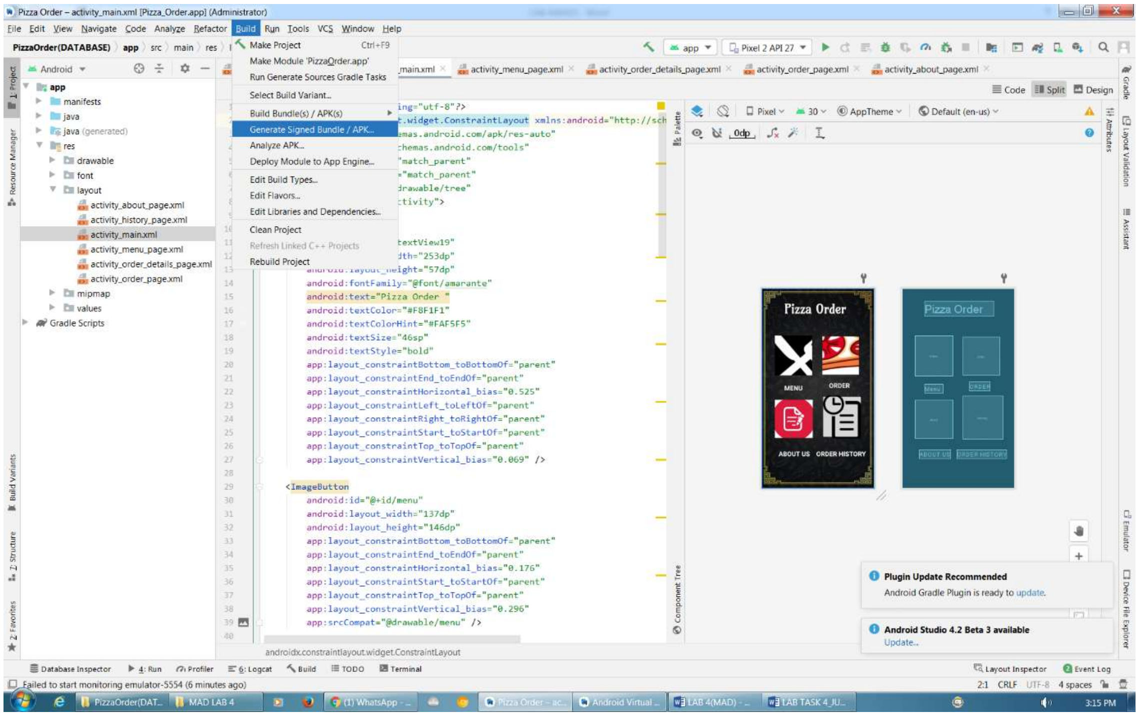
4. Order Tracking Screen

The tracking interface shows real-time order progress with status indicators such as Preparing, Out for Delivery, and Delivered. Maps, driver info, and ETA may also be displayed.

5. Profile & Order History Screen

Users can manage personal details, saved addresses, payment methods, and view previous orders. Ratings and feedback options are also integrated.

1. Signed .apk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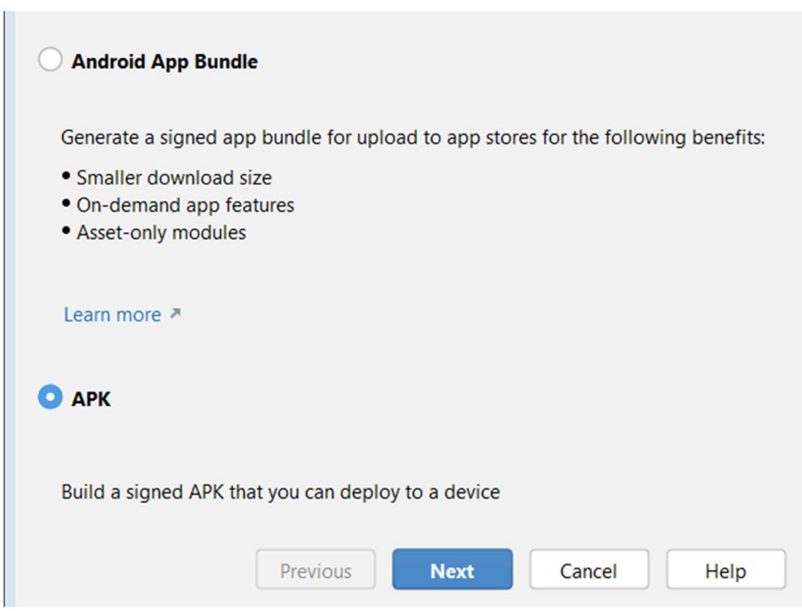
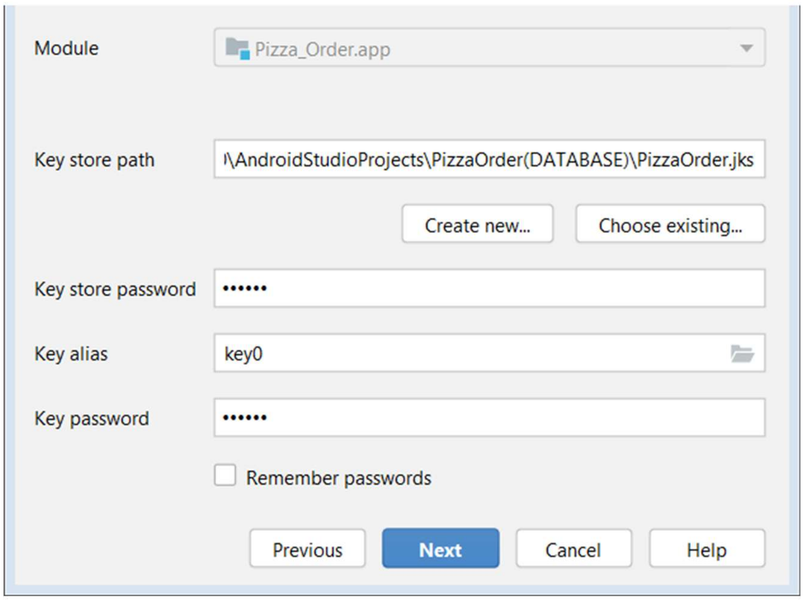
Make Module 'PizzaOrder.app'
Run Generate Sources Gradle Tasks
Select Build Variant...
Build Bundle(s) / APK(s)
Generate Signed Bundle / APK...
Analyze APK...
Deploy Module to App Engine...
Edit Build Types...
Edit Flavors...
Edit Libraries and Dependencies...
Clean Project
Rebuild Project
Pizza Order
Plugin Update Recommended
Android Gradle Plugin is ready to update.
Android Studio 4.2 Beta 3 available


Android App Bundle
Generate a signed app bundle for upload to app stores for the following benefits:
• Smaller download size
• On-demand app features
• Asset-only modules
Learn more ↗
APK
Build a signed APK that you can deploy to a device
Previous
Next
Cancel
Help


Module
Pizza_Order.app
Key store path
\AndroidStudioProjects\PizzaOrder(DATABASE)\PizzaOrder.jks
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Choose existing...
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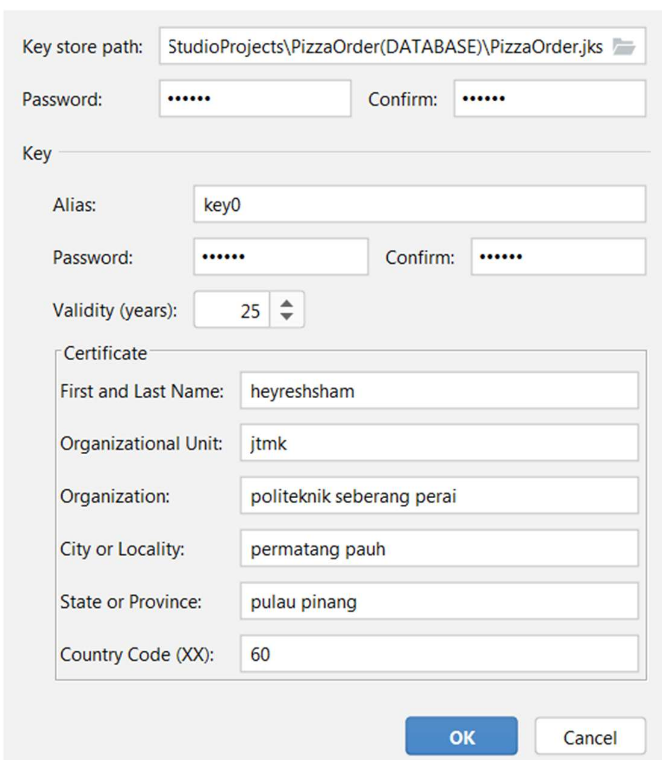
Key alias
key0
Key password
Remember passwords
Previous
Next
Cancel
Help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StudioProjects\PizzaOrder(DATABASE)\PizzaOrder.jks
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Confirm:
Key
Alias:
key0
Password:
Confirm:
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Certificate
First and Last Name:
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Organizational Unit:
jtmk
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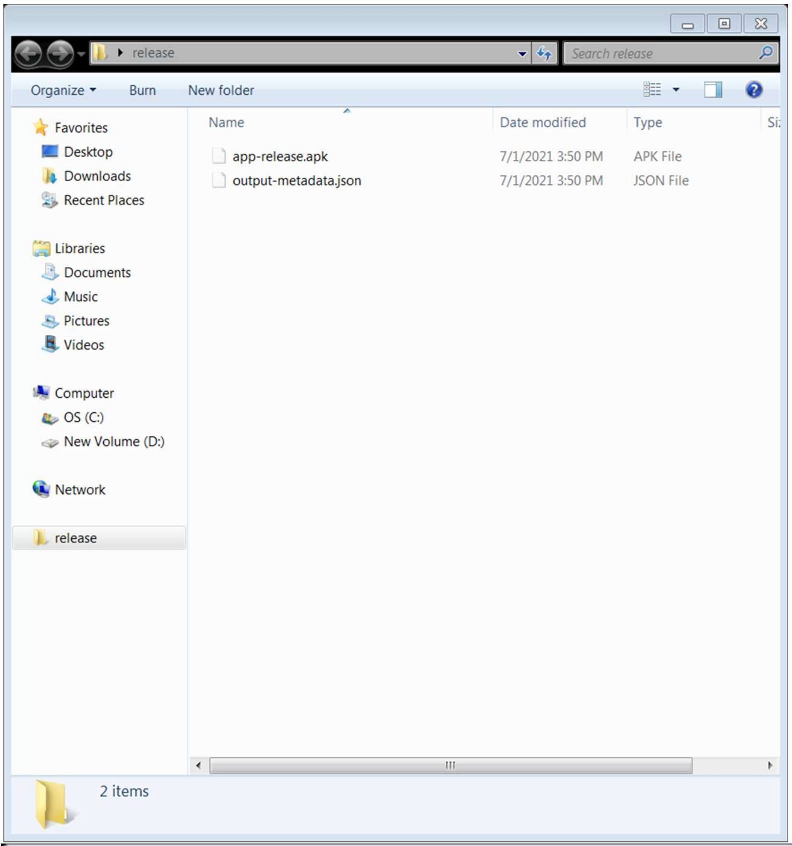
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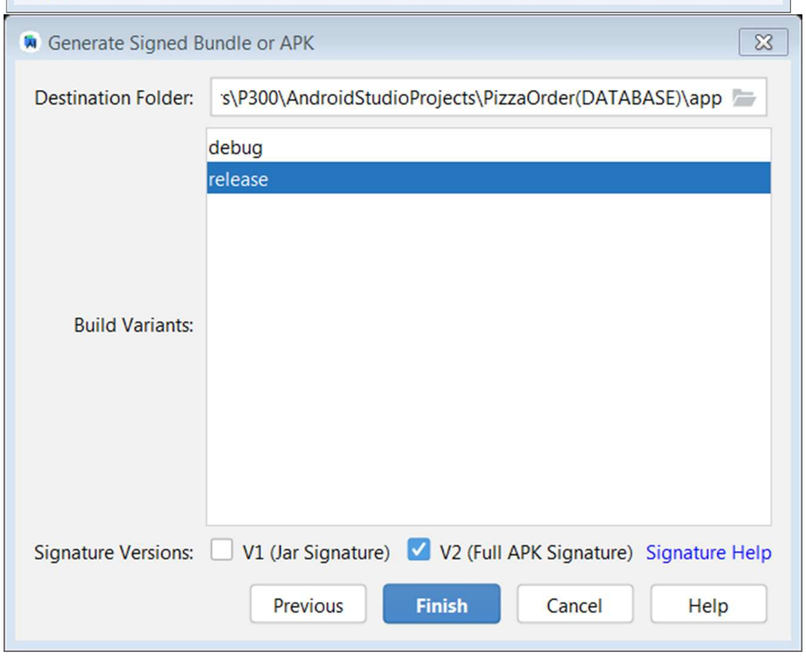
Search release
Organize
Burn
New folder
Favorites
Desktop
Downloads
Recent Places
Libraries
Documents
Music
Pictures
Videos
Computer
OS (C:)
New Volume (D:)
Network
release
Name
Date modified
Type
app-release.apk
7/1/2021 3:50 PM
APK File
output-metadata.json
7/1/2021 3:50 PM
JSON File
2 items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release
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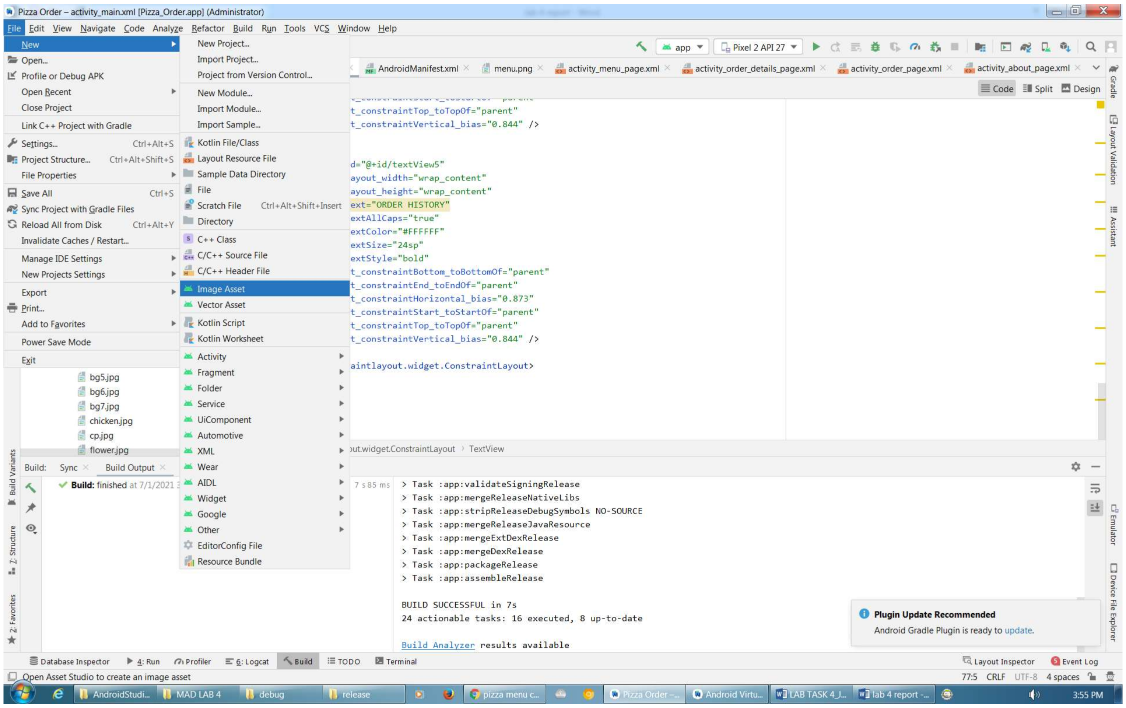
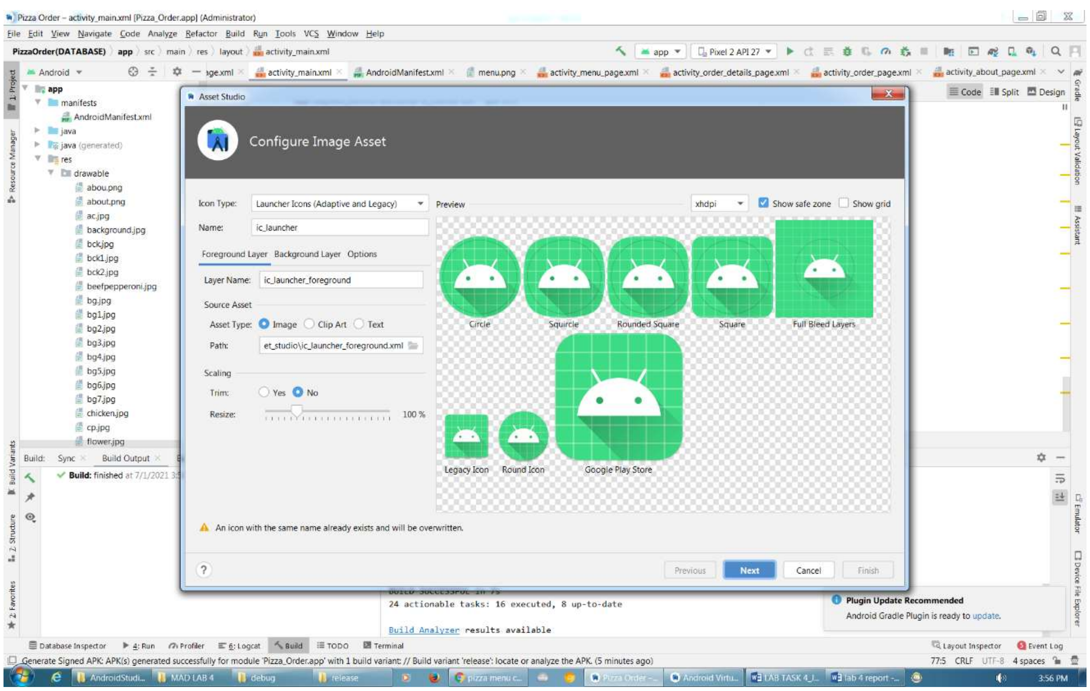
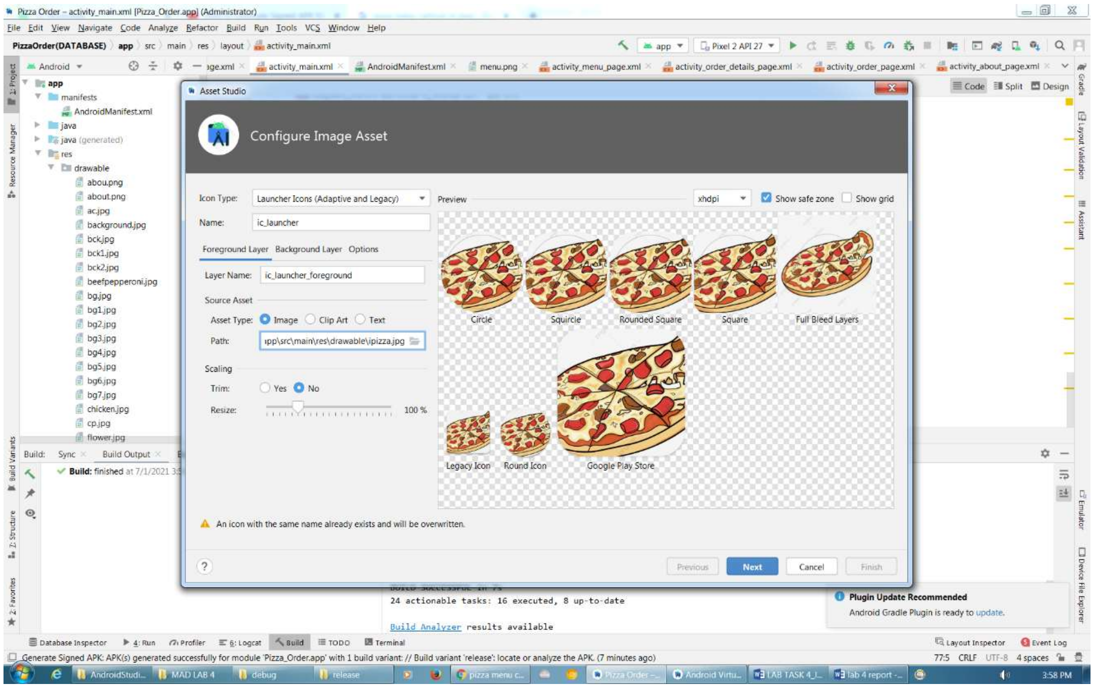
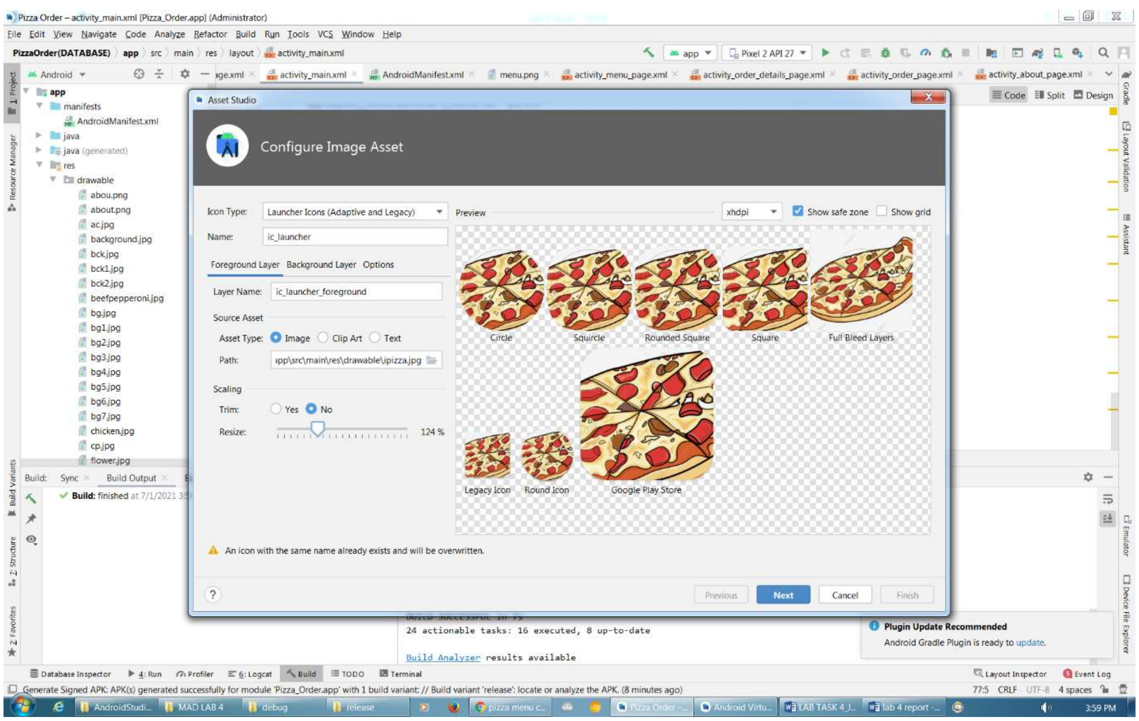
Signature Versions:
V1 (Jar Signature)
V2 (Full APK Signature)
Signature Help
Previous
Finish
Cancel
Help

1. Change Icon






Pizza Order – activity_main.xml [Pizza_Order.app] (Administrator)
Asset Studio
Configure Image Asset
Icon Type: Launcher Icons (Adaptive and Legacy)
Name: ic_launcher
Foreground Layer Background Layer Options
Layer Name: ic_launcher_foreground
Source Asset
Asset Type: Image Clip Art Text
Path: app\src\main\res\drawable\pizza.jpg
Scaling
Trim: Yes No
Resize: 100 %
Preview
xhdpi
Show safe zone
Show grid
Circle
Squircle
Rounded Square
Square
Full Bleed Layers
Legacy Icon
Round Icon
Google Play Store
An icon with the same name already exists and will be overwritten.
Previous
Next
Cancel
Finish
24 actionable tasks: 16 executed, 8 up-to-date
Build Analyzer results available
Plugin Update Recommended
Android Gradle Plugin is ready to update.
Generate Signed APK: APK(s) generated successfully for module 'Pizza_Order.app' with 1 build variant: // Build variant 'release': locate or analyze the APK. (7 minutes ago)
3:58 PM


Pizza Order – activity_main.xml [Pizza_Order.app] (Administrator)
Asset Studio
Configure Image Asset
Icon Type: Launcher Icons (Adaptive and Legacy)
Name: ic_launcher
Foreground Layer Background Layer Options
Layer Name: ic_launcher_foreground
Source Asset
Asset Type: Image Clip Art Text
Path: app\src\main\res\drawable\pizza.jpg
Scaling
Trim: Yes No
Resize: 124 %
Preview
xhdpi
Show safe zone
Show grid
Circle
Squircle
Rounded Square
Square
Full Bleed Layers
Legacy Icon
Round Icon
Google Play Store
An icon with the same name already exists and will be overwritten.
Previous
Next
Cancel
Finish
24 actionable tasks: 16 executed, 8 up-to-date
Build Analyzer results available
Plugin Update Recommended
Android Gradle Plugin is ready to update.
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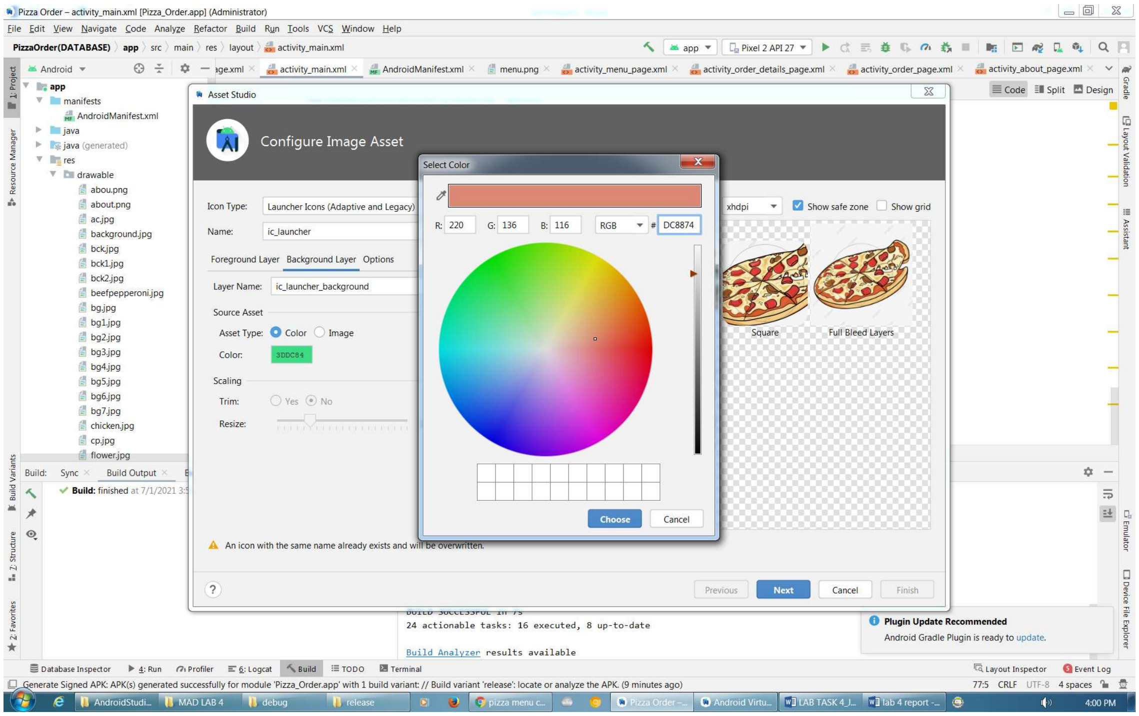
Pizza Order – activity_main.xml [Pizza_Order.app] (Administrator)
Asset Studio
Configure Image Asset
Select Color
R: 220
G: 136
B: 116
RGB
DC8874
Icon Type:
Launcher Icons (Adaptive and Legacy)
Name:
ic_launcher
Foreground Layer
Background Layer
Options
Layer Name:
ic_launcher_background
Source Asset
Asset Type:
Color
Image
Color:
Scaling
Trim:
Yes
No
Resize:
Square
Full Bleed Layers
Choose
Cancel
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Next
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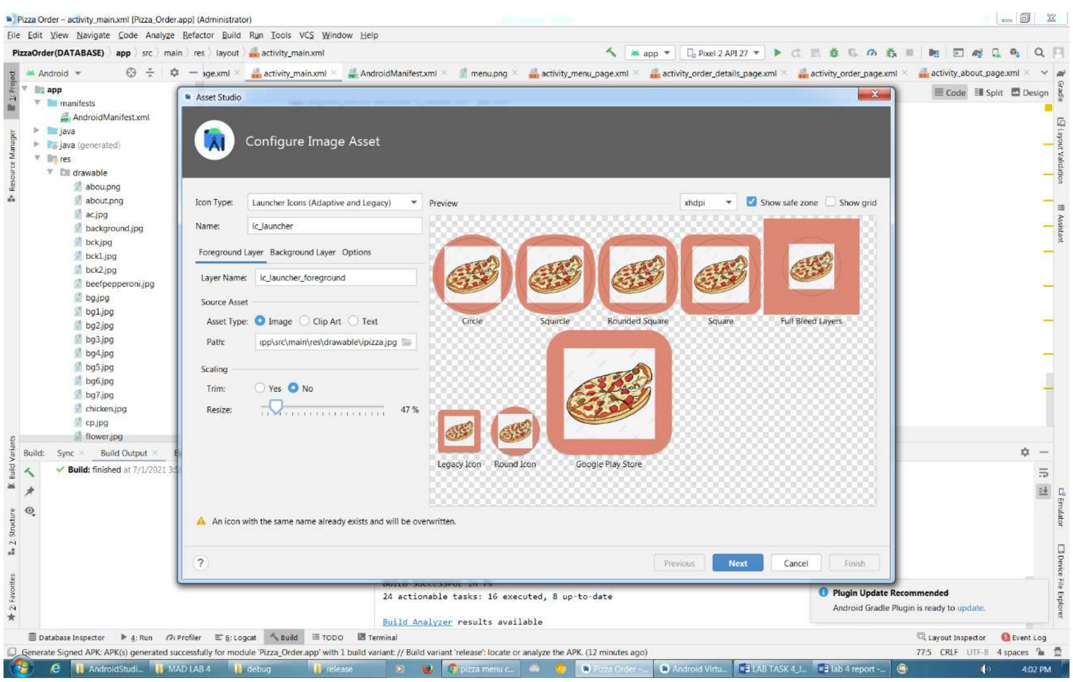
Finish
Plugin Update Recommended
Android Gradle Plugin is ready to update.


Pizza Order – activity_main.xml [Pizza_Order.app] (Administrator)
Asset Studio
Configure Image Asset
Icon Type:
Launcher Icons (Adaptive and Legacy)
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ic_launcher
Foreground Layer
Background Layer
Options
Layer Name:
ic_launcher_foreground
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Asset Type:
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Clip Art
Text
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Scaling
Trim:
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Resize:
47 %
Preview
xhdpi
Show safe zone
Show grid
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Square
Full Bleed Layers
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Google Play Store
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Plugin Update Recommended
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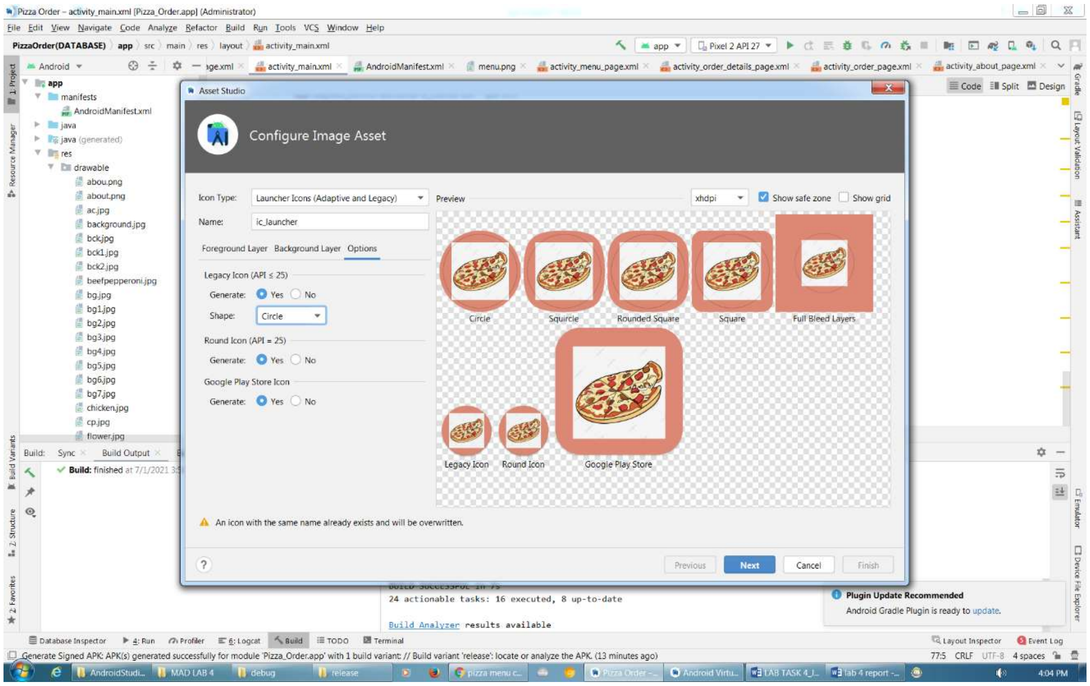
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Generate: Yes No
Google Play Store Icon
Generate: Yes No
Preview
Circle
Squircle
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Square
Full Bleed Layers
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Round Icon
Google Play Store
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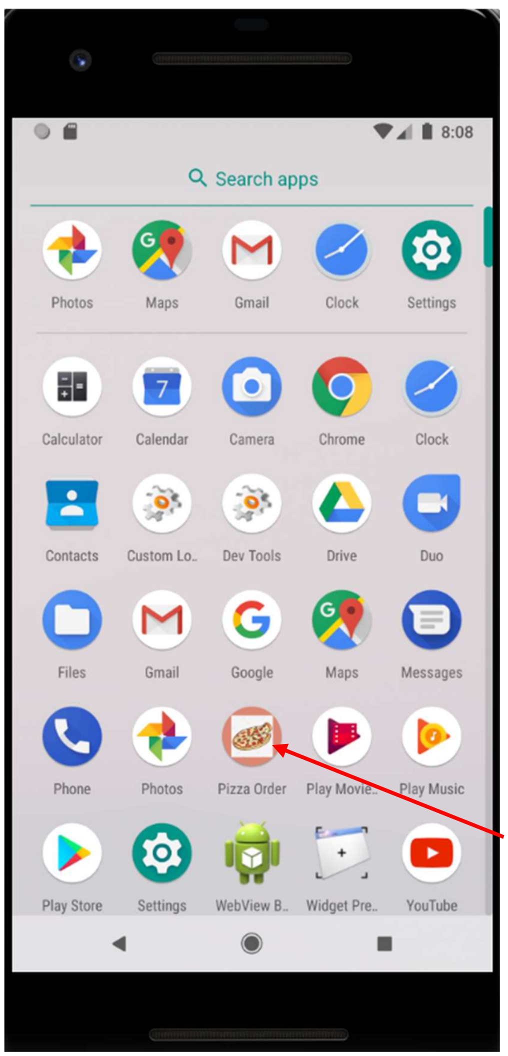
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Photos
Maps
Gmail
Clock
Settings
Calculator
Calendar
Camera
Chrome
Clock
Contacts
Custom Lo..
Dev Tools
Drive
Duo
Files
Gmail
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Messages
Phone
Photos
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Play Movie..
Play Music
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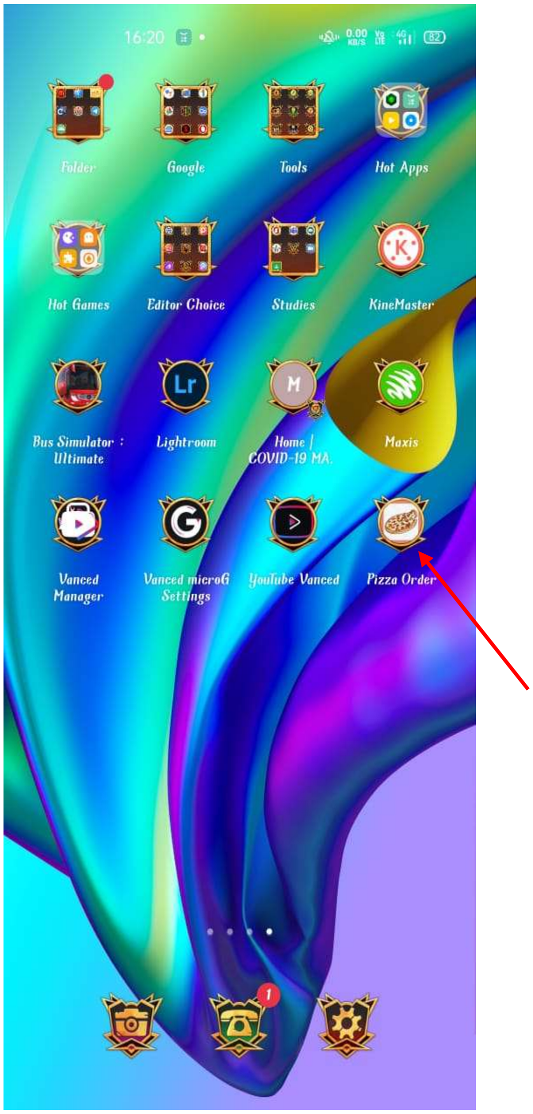
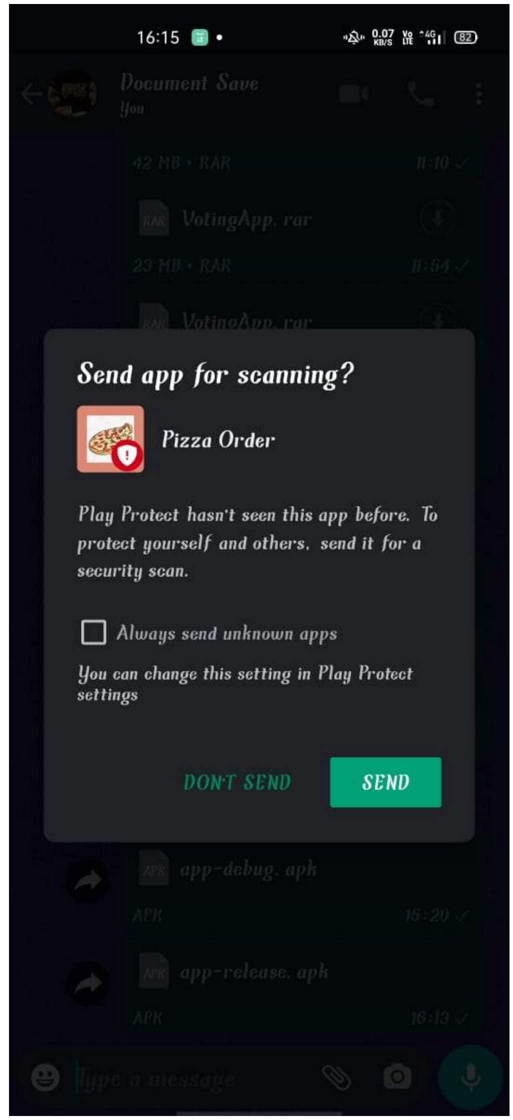
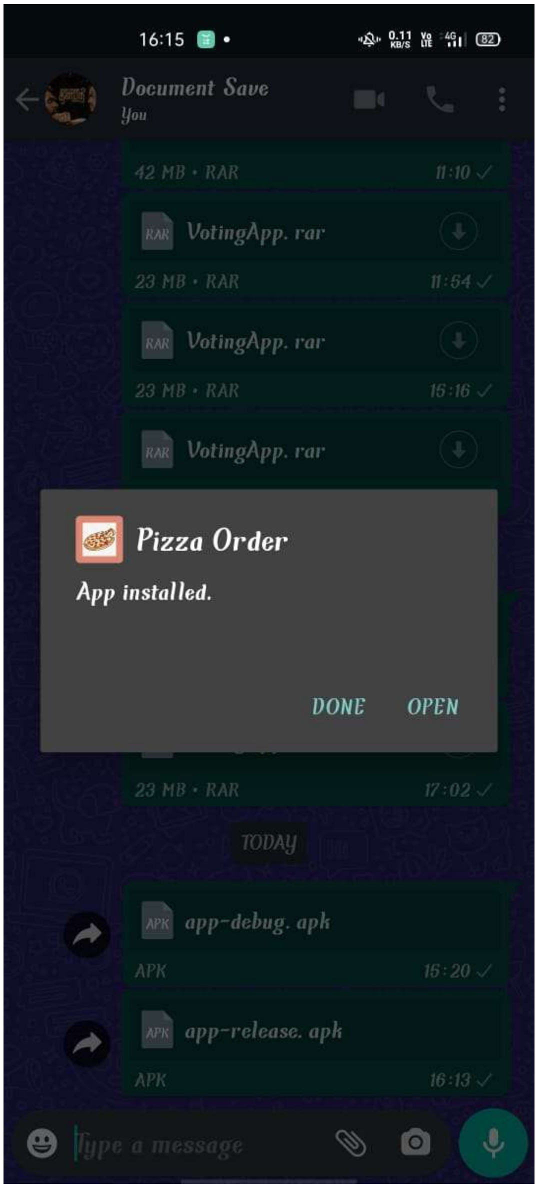
Settings
WebView B..
Widget Pre..
YouTube

1. Installation



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You
42 MB • RAR 11:10
VotingApp. rar
23 MB • RAR 11:54
VotingApp. rar
23 MB • RAR 16:16
VotingApp. rar
Pizza Order
App installed.
DONE
OPEN
23 MB • RAR 17:02
TODAY
app-debug. apk
APK 16:20
app-release. apk
APK 16:13
Type a message


16:15
Pizza Order
Pizza
MENU
ORDER
ABOUT US
ORDER HISTORY


16:15
Pizza Order
Pizza Ordering
MENU
PIzza Chicken Pepperoni
Loaded with extra pepperoni and extra cheese
Hawaiian Chicken
with tomato sauce, beef pepperoni, mozzarella cheese
Aloha Chicken
with thousand island sauce, tuna, pineapples, onions, mozzarella cheese.
BACK


16:16
Pizza Order
Pizza Ordering
ORDER
Pizza Menu
Size:
Small - RM5
Medium -RM10
Large - RM15
Topping:
Seafood - RM8
Chicken - RM5
Meat - RM6
Heyreshsham
perak
019373788
BIGSTOCK
ORDER
bigstock.com


16:16
Pizza Order
Pizza Ordering
ORDER
PIZZA
Your Order :
Pizza Size : Small - RM5
Pizza Topping : Chicken - RM5
Heyreshsham
perak
019373788
Total Price: RM :10.0
Data Inserted
BACK
MAIN


16:16
Pizza Order
Pizza Ordering
ABOUT US
Jalan Tok Nasir
TAMAN
PAUH INDAH
TAMAN
UH JAYA
Jalan Pauh Jaya
Jalan Zamrud
Jalan Berlian
CANADIAN PIZZA
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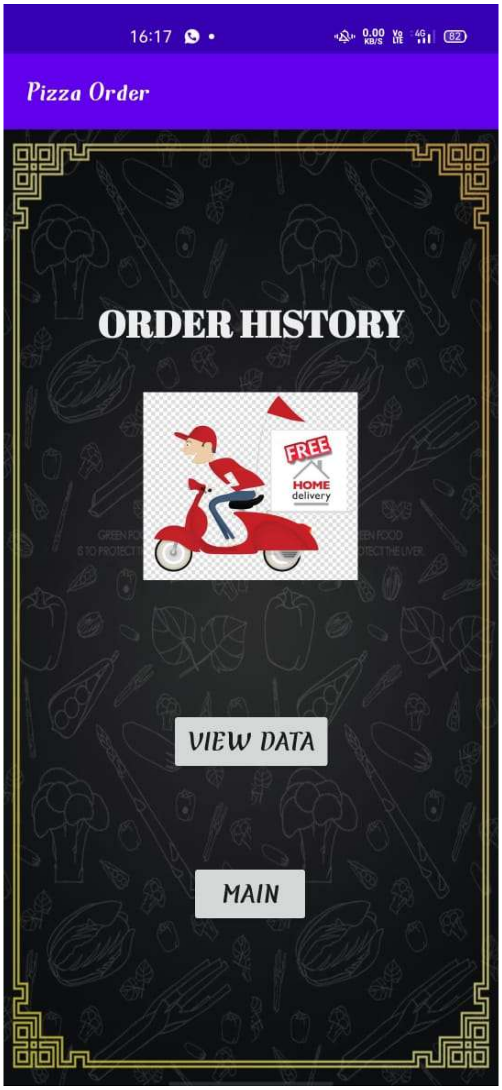
TEL:1-300-88-2525
HOURS: 11:00AM TO 11:00PM
BACK


16:17
Pizza Order
ORDER HISTORY
FREE
HOME
delivery
VIEW DATA
MAIN


16:17
Pizza Order
ORDER HISTORY
FREE
Data
Id :1
CustName : Heyreshsham
VIEW DATA
MAIN

5.0 Design Alignment User Needs

5.1 User Need Analysis

User Needs Analysis identifies the expectations, problems, and requirements of users who interact with the Pizza Ordering App. Understanding these needs ensures the system is designed to provide convenience, efficiency, and satisfaction for all user groups.

Firstly, **customers** need a simple and intuitive platform to order pizzas quickly. They require easy menu browsing, clear images, detailed descriptions, customization options (size, toppings, crust), and transparent pricing. Customers also expect secure payment methods, real-time order tracking, delivery time estimates, and the ability to save addresses and reorder previous meals.

Secondly, **restaurant admins and staff** need efficient tools to manage daily operations. Their needs include menu management (add, edit, delete pizzas), price updates, promotion settings, and inventory visibility. They also require real-time order notifications, order status update features, and dashboards to monitor sales and performance.

Thirdly, **delivery staff** require features that streamline the delivery process. They need instant notifications for assigned deliveries, access to customer addresses and contact details, route navigation support, and the ability to update delivery statuses such as Picked Up, On the Way, and Delivered.

From a business perspective, **business owners** need reporting and analytics capabilities. They require access to sales reports, peak order times, popular menu items, and customer feedback to support decision-making and business growth strategies.

Finally, all users share common system needs such as fast app performance, data security, reliable notifications, and minimal downtime. Addressing these needs ensures the Pizza Ordering App delivers a seamless, secure, and user-friendly experience for customers, staff, and management alike.

5.2 Design Rationale and Justification

The design rationale for the Pizza Ordering App is based on creating a system that is user-friendly, scalable, secure, and efficient for handling online food ordering operations. Each design decision was made to enhance usability, streamline business processes, and ensure the application can support real-time interactions between customers, restaurant staff, and delivery personnel.

Firstly, the app adopts a layered architecture (Presentation, Business Logic, Data, and Core layers) to ensure separation of concerns. This structure improves maintainability, allowing developers to modify the user interface, business rules, or database integration independently without affecting the entire system. It also supports scalability as new features such as loyalty programs or new payment gateways can be added easily.

Secondly, the BLoC (Business Logic Component) pattern is implemented for state management. This approach ensures predictable state transitions, organized event handling, and improved testability. By separating UI from business logic, the app maintains clean code practices and simplifies debugging, especially when managing complex processes like cart updates, order tracking, and authentication flows.

The system integrates Firebase services (Authentication, Firestore, Storage, Messaging) as the backend solution. Firebase Authentication ensures secure login and user management, while Firestore provides real-time database synchronization for orders and menu updates. Firebase Storage manages pizza images, and Firebase Messaging enables push notifications for order updates. This cloud-based approach reduces infrastructure costs and supports real-time performance.

From a user experience perspective, the UI follows modern mobile design principles such as bottom navigation, card-based layouts, high-resolution food imagery, and simple checkout flows. These choices reduce user effort, improve navigation speed, and increase customer engagement, ultimately encouraging repeat orders.

Finally, security and performance considerations influenced the design. Role-based access control ensures only authorized users manage menus or orders, while optimized Firestore queries and lazy loading improve response times. Overall, the design decisions collectively justify a system that is robust, scalable, secure, and aligned with industry best practices for mobile food ordering applications.

6.0 Conclusion

In conclusion, the development of the Pizza Ordering App demonstrates the importance of user-centered design, functional efficiency, and visual usability in mobile application development. Through the user needs analysis, key requirements such as easy menu browsing, customizable pizza selection, secure payment options, and real-time order tracking were identified as essential features to enhance customer satisfaction.

The low-fidelity and high-fidelity UI designs played a significant role in visualizing the system workflow and interface structure. Low-fidelity sketches helped in planning layout placement and navigation flow, while high-fidelity designs refined the visual presentation, color schemes, typography, and interactive components to deliver a more realistic user experience.

Furthermore, the design rationale justified each interface decision based on usability principles such as simplicity, consistency, accessibility, and responsiveness. These considerations ensure that the app is user-friendly for a wide range of customers, including first-time users.

Overall, the Pizza Ordering App provides a convenient digital solution for customers to place orders efficiently while supporting business operations through organized order management. Future enhancements may include AI-based recommendations, loyalty rewards programs, and integration with delivery tracking systems to further improve service quality and user engagement.